Student Handout

Name: ______________________ Date: ______________ Class: ____________

MyDesign Portfolio

Element E Initial Template

Element E: Application of STEM Principles and Practices

STEM Principles and Practices to Design Requirements *(E1)*

STEM Principle for High-Priority Design Requirement #1
STEM Concept:

Friction(Science): A doorstop increases friction between the floor and the bottom of the door.

Describe how STEM concept applies directly to the specifics of your design:

Stem concept applies to this part because friction is physics which is a type of science.

STEM Principle for High-Priority Design Requirement #2
STEM Concept:

Doorstop(Technology): The door stop is a simple tool that solves a common problem: keeping a door open or closed

Describe how STEM concept applies directly to the specifics of your design:

Different materials (rubber, plastic, metal) are used to improve grip, durability, and effectiveness.

STEM Principle for High-Priority Design Requirement #3
STEM Concept:

Name: ______________________________ **Date:** _________________ **Class:** _______________

Wedge (Engineering): The door stop is a wedge, one of the six classical simple machines.

Describe how STEM concept applies directly to the specifics of your design:

Engineering design door stops with the right angle, shape, and material to create enough resistance and be easy to use.

Proposed Solution to STEM Principles and Practices *(E2)*

STEM Principle for Design Solution #1
STEM Concept:

Friction principle of science.

Describe how STEM concept applies directly to the specifics of your solution:

Helps the big problem of the door stop sliding from the door and not working.

STEM Principle for Design Solution #2
STEM Concept:

Door stop princleble of Technology

Describe how STEM concept applies directly to the specifics of your solution:

Name: ______________________________ **Date:** _________________ **Class:** _______________

Technology is the T in STEM. The door stop is a type of door stopper that utilizes the friction and weight of a rubber or cushioned material to hold the door open.

STEM Principle for Design Solution #3

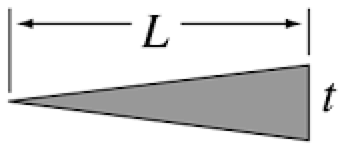
STEM Concept:

Wedge: the E in Stem

Describe how STEM concept applies directly to the specifics of your solution:

A wedge engineering, in its simplest form, functions by converting a force applied to its base (or thick end) into a force directed along its inclined surfaces (or thin edge)

Equations being used used (E3)



Wedge $IMA = \frac{L}{t}$

L = depth of penetration

t = separation of wedged surfaces

friction force

normal force

$$f_{friction} = \mu N$$

coefficient of friction

Name: ______________________________ **Date:** _________________ **Class:** _______________